PHASE 8

Power BI DAX & Dashboard

KPI measures, time intelligence, interactivity, visual design, and final dashboard QA

B2B Travel Booking & Revenue Analytics

Complete Understanding Notes | Beginner-Friendly | Portfolio & Interview Ready

Purpose: understand what was done, why it was done, how it was validated, and how to explain it in an interview.

Prepared from the completed project work and verified analysis results.

Contents and Learning Map

Use this document in two ways: first as a learning summary, then as an interview revision sheet. Each section explains the concept, implementation, validation, and business meaning.

#	Section	What you should understand
1	Dashboard objective and KPI layer	What the dashboard must communicate and which KPIs are primary.
2	Core DAX measures	Booking, revenue, margin, cancellations, and refunds.
3	Time intelligence	Previous-month and MoM measures plus date sorting.
4	Visual design	Trend, top-N, cancellation, status distribution, and slicers.
5	Conditional formatting and UX	Positive/negative growth colors, Data Period, and Clear Filters.
6	Validation and dashboard interpretation	Cross-checks against Python/SQL and the final KPI story.
7	Interview revision and final checklist	How to explain the dashboard as an analyst.

Project rule: Do not memorize code alone. Be able to explain the grain of the data, why each step is needed, and how you verified the result.

1. Dashboard Objective and KPI Layer

The Phase 8 dashboard is designed for fast management scanning: first understand scale and profitability, then see the monthly trend, then identify top commercial contributors, and finally examine cancellations and refunds.

KPI	Definition	Verified full-data result
Total Bookings	Count of valid booking fact rows	119,877
Total GBV	Sum of gross_booking_value_inr	about INR 26.15B
Net Revenue	Sum of net_revenue_inr	about INR 3.25B
Average Booking Value	GBV / Total Bookings	about INR 218.15K
Gross Margin %	Net Revenue / GBV x 100	12.43%
Booking Cancellation Rate	Cancelled Bookings / Total Bookings x 100	about 11.64%
Cancellation Events	Rows in cancellation fact	13,972
Total Refund	Sum of refund_amount_inr	about INR 2.93B

Important distinction: Cancelled Bookings and Cancellation Events are not identical. The business cancellation-rate KPI should use cancelled booking rows (13,959), not cancellation-event rows (13,972).

2. Core DAX Measures

```
Total Bookings = COUNTROWS(analytics_fact_bookings_fx)

Total GBV INR =
SUM(analytics_fact_bookings_fx[gross_booking_value_inr])

Total Net Revenue INR =
SUM(analytics_fact_bookings_fx[net_revenue_inr])

Average Booking Value INR =
DIVIDE([Total GBV INR], [Total Bookings])

Gross Margin % =
DIVIDE([Total Net Revenue INR], [Total GBV INR]) * 100
```

```
Cancellation Events =
COUNTROWS(analytics_fact_cancellations_fx)

Cancelled Bookings =
CALCULATE(
    [Total Bookings],
    analytics_fact_bookings_fx[booking_status] = "Cancelled"
)

Cancellation Rate % =
DIVIDE([Cancelled Bookings], [Total Bookings]) * 100

Total Refund INR =
SUM(analytics_fact_cancellations_fx[refund_amount_inr])
```

Optional display measures

```
Total GBV B = DIVIDE([Total GBV INR], 1000000000)
Total Net Revenue B = DIVIDE([Total Net Revenue INR], 1000000000)
ABV K = DIVIDE([Average Booking Value INR], 1000)
```

These scaled measures are useful when card formatting options do not expose the desired display units. The underlying business logic remains unchanged.

3. Time Intelligence and MoM Growth

```
Year Month = FORMAT(Date_Table[Date], "YYYY-MM")
Year Month Sort = YEAR(Date_Table[Date]) * 100 + MONTH(Date_Table[Date])
```

Year Month should be sorted by Year Month Sort so the axis stays chronological.

```
Previous Month GBV =
CALCULATE([Total GBV INR], DATEADD(Date_Table[Date], -1, MONTH))

GBV MoM Growth % =
DIVIDE([Total GBV INR] - [Previous Month GBV], [Previous Month GBV]) * 100

Previous Month Bookings =
CALCULATE([Total Bookings], DATEADD(Date_Table[Date], -1, MONTH))

Booking MoM Growth % =
DIVIDE([Total Bookings] - [Previous Month Bookings], [Previous Month Bookings]) * 100

Previous Month Net Revenue =
CALCULATE([Total Net Revenue INR], DATEADD(Date_Table[Date], -1, MONTH))

Net Revenue MoM Growth % =
DIVIDE([Total Net Revenue INR] - [Previous Month Net Revenue], [Previous Month Net Revenue]) * 100
```

Validation point	Power BI result	Python result	Status
Feb 2024 GBV MoM	-8.68%	-8.68%	Matched
Dec 2024 GBV MoM	+11.31%	+11.31%	Matched

4. Dashboard Visuals and What They Answer

Visual	Fields / measure	Business question
KPI cards	Bookings, GBV, Net Revenue, ABV, Margin, Cancellation Rate, Refunds	How big, profitable, and cancellation-exposed is the business?
Monthly trend	Year Month + GBV + Net Revenue	How do value and revenue move over time?
GBV MoM growth	Year Month + GBV MoM Growth %	Which months accelerated or contracted?
Top 10 destinations	destination_id + Total GBV	Where is booking value concentrated?
Top 10 suppliers	supplier_id + Net Revenue	Which suppliers contribute most revenue?
Top 10 hotels	hotel_id + Net Revenue	Which hotels contribute most revenue?
Top 10 agents	agent_id + Net Revenue	Which agents are most valuable?
Booking status donut	booking_status + Total Bookings	What is the booking-status mix?
Cancellations by reason	cancellation_reason + event count	Why are bookings cancelled?
Refund by reason	cancellation_reason + Total Refund	Which reasons drive refund exposure?

Slicer panel

The left-side slicer panel uses separate slicers for Year, Booking Status, Agent, Country, Hotel, and Supplier. Keeping them separate avoids hierarchy mapping errors and gives users predictable filtering behavior.

5. Conditional Formatting and Dashboard UX

Positive/negative MoM colors

The GBV MoM chart uses rule-based conditional formatting so growth and decline are visually distinct.

```
Rule 1: value >= -100 and < 0   -> Red
Rule 2: value >= 0 and < 100    -> Green
```

The DAX measure already returns percentage-point values because it multiplies by 100, so numeric rules such as -100 to 100 are appropriate for this dataset.

Data Period measure

```
Data Period =
VAR MinDate =
    MINX(ALLSELECTED(Date_Table), Date_Table[Date])
VAR MaxDate =
    MAXX(ALLSELECTED(Date_Table), Date_Table[Date])
RETURN
    FORMAT(MinDate, "MMM yyyy")
    & " - "
    & FORMAT(MaxDate, "MMM yyyy")
```

Clear Filters bookmark pattern

1. Reset all slicers to All.
2. Create a bookmark named Clear Filters and keep Data enabled.
3. Insert a button and set Action = Bookmark.
4. Point the button to the Clear Filters bookmark.
5. Test by applying several slicers and then clicking the button.

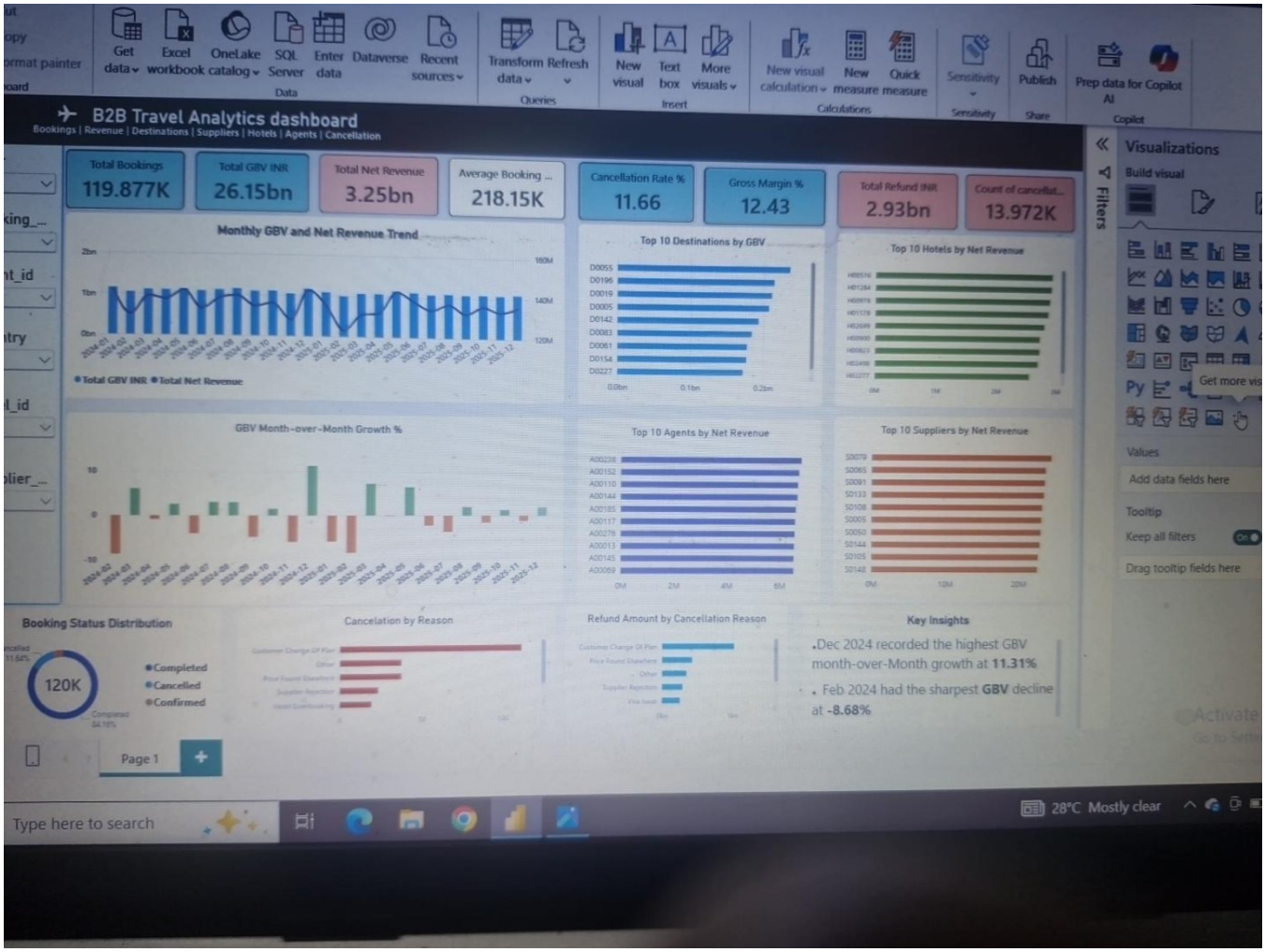
6. Booking Status Distribution and Key Insights Panel

The booking-status donut uses the three real status categories in the project: Completed, Cancelled, and Confirmed. It should not copy extra categories from a visual reference if those categories do not exist in the dataset.

```
Legend: analytics_fact_bookings_fx[booking_status]
Values: [Total Bookings]
```

A Key Insights text box turns the dashboard from a collection of charts into a management story. Suitable verified bullets include Dec 2024 +11.31% GBV MoM, D0055 as top destination, S0079 as top supplier, H00516 as top hotel by net revenue, A00238 as top agent by net revenue, and Customer Change Of Plan as the leading cancellation/refund driver.

7. Final Dashboard Snapshot and QA



Final dashboard development snapshot from the project.

QA note: Some dashboard screenshots during development displayed approximately 11.66% because event count was used in the cancellation-rate calculation. The final business KPI should use Cancelled Bookings / Total Bookings, giving about 11.64%.

Cross-tool validation

Metric / finding	Expected full-data result
Total Bookings	119,877
Total GBV	about INR 26.15B
Net Revenue	about INR 3.25B
ABV	about INR 218.15K
Gross Margin	12.43%
Cancellation Rate	about 11.64%
Feb 2024 GBV MoM	-8.68%
Dec 2024 GBV MoM	+11.31%

8. Interview Revision and Phase 8 Completion

60-second dashboard explanation: I built a Power BI dashboard on top of the validated semantic model using DAX measures for bookings, GBV, net revenue, average booking value, margin, cancellation rate, refunds, and MoM growth. I validated time-intelligence values against Python, added slicers and a reset pattern for usability, used conditional formatting to distinguish positive and negative growth, and designed the page so executives can move from KPIs to trends, top partners, and cancellation/refund drivers. I also separated booking cancellation rate from cancellation-event count to keep the KPI definition consistent.

Phase 8 completion checklist

- Core DAX measures created.
- MoM measures validated against Python.
- KPI cards and monthly trend created.
- Top destination/supplier/hotel/agent visuals created.
- Cancellation and refund visuals created.
- Booking status donut created.
- Slicers and conditional formatting configured.
- Key Insights panel added.
- Final KPI definitions reviewed for consistency.